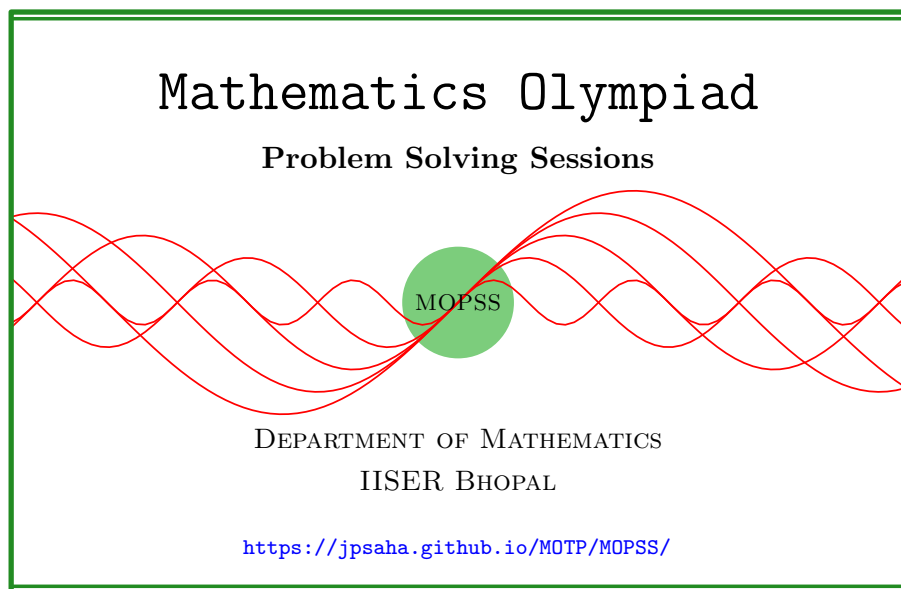


Growth of polynomials

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

List of problems and examples

1.1	Example (India BStat-BMath 2012)	2
1.2	Example	3
1.3	Example	4
1.4	Example (India RMO 2015c P3)	4
1.5	Example	6

§1 On the growth of polynomials

Example 1.1 (India BStat-BMath 2012). Show that the polynomial $x^8 - x^7 + x^2 - x + 15$ has no real root.

Summary — In absolute value, a higher degree polynomial dominates a smaller degree polynomial at arguments which are large enough in absolute value.

Solution 1. Let α be a real number. Let us consider the following cases.

1. $\alpha \geq 1$,
2. $\alpha \leq 0$,
3. $0 \leq \alpha \leq 1$.

If $\alpha \geq 1$, then

$$\begin{aligned} \alpha^8 - \alpha^7 + \alpha^2 - \alpha + 15 &= \alpha^7(\alpha - 1) + \alpha(\alpha - 1) + 15 \\ &\geq 15. \end{aligned}$$

If $\alpha \leq 0$, then

$$\begin{aligned} \alpha^8 - \alpha^7 + \alpha^2 - \alpha + 15 &= \alpha^8 + (-\alpha^7) + \alpha^2 + (-\alpha) + 15 \\ &\geq 15. \end{aligned}$$

If $0 \leq \alpha \leq 1$, then

$$\begin{aligned} \alpha^8 - \alpha^7 + \alpha^2 - \alpha + 15 &= \alpha^8 + (1 - \alpha^7) + \alpha^2 + (1 - \alpha) + 13 \\ &\geq 13. \end{aligned}$$

It follows that the polynomial $x^8 - x^7 + x^2 - x + 15$ has no real root. ■

Example 1.2. Does there exist a polynomial $P(x)$ with rational coefficients such that $\sin x = P(x)$ for all $x \geq 100$?

Solution 2. Suppose there exists a polynomial $P(x)$ with rational coefficients such that $\sin x = P(x)$ for all $x \geq 100$. It follows that $P(x)$ has absolute value at most 1 for all $x \geq 100$.

Claim — Let $f(x)$ be a nonconstant polynomial with real coefficients. Then for any given $M > 0$, there exists a real number $x_0 > 0$ such that

$$|f(x)| > M$$

for all real number x satisfying $|x| > x_0$.

Proof of the Claim. Write

$$f(x) = a_d x^d + a_{d-1} x^{d-1} + \cdots + a_1 x + a_0,$$

where $a_d, a_{d-1}, \dots, a_0$ lie in $\mathbb{R}$ and d denotes the degree of $f(x)$. Note that for any real α ,

$$\begin{aligned} |f(\alpha)| &= |a_d \alpha^d + a_{d-1} \alpha^{d-1} + \cdots + a_1 \alpha + a_0| \\ &\geq |a_d \alpha^d| - |a_{d-1} \alpha^{d-1} + \cdots + a_1 \alpha + a_0| \\ &\geq |a_d \alpha^d| - |a_{d-1} \alpha^{d-1}| - \cdots - |a_1 \alpha| - |a_0| \\ &= \left(\frac{1}{d} |a_d \alpha^d| - |a_{d-1} \alpha^{d-1}| \right) \\ &\quad + \left(\frac{1}{d} |a_d \alpha^d| - |a_{d-2} \alpha^{d-2}| \right) \\ &\quad + \cdots + \left(\frac{1}{d} |a_d \alpha^d| - |a_1 \alpha| \right) \\ &\quad + \left(\frac{1}{d} |a_d \alpha^d| - |a_0| \right) \\ &= \frac{|a_d|}{d} |\alpha^{d-1}| \left(|\alpha| - \left| \frac{da_{d-1}}{a_d} \right| \right) \\ &\quad + \frac{|a_d|}{d} |\alpha^{d-2}| \left(|\alpha|^2 - \left| \frac{da_{d-2}}{a_d} \right| \right) \\ &\quad + \cdots + \frac{|a_d|}{d} |\alpha| \left(|\alpha|^{d-1} - \left| \frac{da_1}{a_d} \right| \right) \\ &\quad + \frac{|a_d|}{d} \left(|\alpha|^d - \left| \frac{da_0}{a_d} \right| \right). \end{aligned}$$

Hence, for any given $M > 0$,

$$|f(x)| > M$$

holds for any real number x of large enough absolute value. Indeed, for any real number x satisfying

$$\begin{aligned} |x| &> \left| \frac{da_{d-1}}{a_d} \right|, \\ |x|^2 &> \left| \frac{da_{d-2}}{a_d} \right|, \\ &\dots > \dots, \\ |x|^{d-1} &> \left| \frac{da_1}{a_d} \right|, \\ |x|^d &> \left| \frac{da_0}{a_d} \right| + \frac{Md}{|a_d|}, \end{aligned}$$

or equivalently, satisfying

$$x > \max \left\{ \left| \frac{da_{d-1}}{a_d} \right|, \left(\left| \frac{da_{d-2}}{a_d} \right| \right)^{1/2}, \dots, \left(\left| \frac{da_1}{a_d} \right| \right)^{1/(d-1)}, \left(\left| \frac{da_0}{a_d} \right| + \frac{Md}{|a_d|} \right)^{1/d} \right\},$$

the inequality $|f(x)| > M$ holds. The Claim follows by taking

$$x_0 = \max \left\{ \left| \frac{da_{d-1}}{a_d} \right|, \left(\left| \frac{da_{d-2}}{a_d} \right| \right)^{1/2}, \dots, \left(\left| \frac{da_1}{a_d} \right| \right)^{1/(d-1)}, \left(\left| \frac{da_0}{a_d} \right| + \frac{Md}{|a_d|} \right)^{1/d} \right\}.$$

□

By the above Claim, it follows that $P(x)$ is a constant polynomial. This shows that $\sin x$ is constant on the interval $[100, \infty)$, which is impossible since $\sin 100\pi \neq \sin 101\pi$ and $101\pi, 100\pi$ lie in $[100, \infty)$. This contradicts the assumption that there exists a polynomial $P(x)$ with rational coefficients such that $\sin x = P(x)$ for all $x \geq 100$.

Hence, there does not exist a polynomial $P(x)$ with rational coefficients such that $\sin x = P(x)$ for all $x \geq 100$. ■

Example 1.3. Show that a polynomial function from $\mathbb{R}$ to $\mathbb{R}$ can be expressed as the difference of two strictly increasing polynomial functions in infinitely many ways.

Example 1.4 (India RMO 2015c P3). Let $P(x)$ be a nonconstant polynomial whose coefficients are positive integers. If $P(n)$ divides $P(P(n) - 2015)$ for all natural numbers n , then prove that $P(-2015) = 0$.

Summary — In absolute value, a higher degree polynomial dominates a smaller degree polynomial at arguments which are large enough in absolute value.

Solution 3. Let $Q(x), R(x)$ be polynomials with rational coefficients such that

$$P(P(x) - 2015) = P(x)Q(x) + R(x)$$

and $R(x) = 0$ or $\deg R(x) < \deg P(x)$. Note that $P(n)$ is positive for all integer $n \geq 1$ since the coefficients of $P(x)$ are positive integers. By the given condition, it follows that $P(n)$ divides $R(n)$ for any integer $n \geq 1$.

Claim — Let $f(x), g(x)$ be two polynomials with real coefficients. Suppose $f(x)$ is a nonconstant polynomial with a positive leading coefficient, and $\deg g(x) < \deg f(x)$. Then there exists an integer $n_0 \geq 1$ such that

$$f(n) > g(n)$$

for any $n \geq n_0$.

Proof of the Claim. Note that it suffices to prove the Claim if $f(x)$ is a monomial, that is, a power of x . Indeed, write $f(x) = a_d x^d + a_{d-1} x^{d-1} + \cdots + a_0$ with $a_0, \dots, a_d \in \mathbb{R}$ and d denoting the degree of f . Also write $g(x) = b_e x^e + b_{e-1} x^{e-1} + \cdots + b_0$ with $b_0, \dots, b_e \in \mathbb{R}$ and e denoting the degree of g . Noting that $a_d > 0$, it follows that for a positive integer n , the inequality

$$a_d n^d + a_{d-1} n^{d-1} + \cdots + a_0 > b_e n^e + b_{e-1} n^{e-1} + \cdots + b_0$$

holds if

$$a_d n^d > \frac{a_{d-1}}{a_d} n^{d-1} + \cdots + \frac{a_0}{a_d} + \frac{b_e}{a_d} n^e + \frac{b_{e-1}}{a_d} n^{e-1} + \cdots + \frac{b_0}{a_d}$$

is satisfied, which can be concluded provided the Claim is known in the case when f is a monomial.

Let us assume that f is a monomial. Write $f(x) = x^d$ where d denotes the degree of f , and write $g(x) = b_e x^e + b_{e-1} x^{e-1} + \cdots + b_0$ with $b_0, \dots, b_e \in \mathbb{R}$ and e denoting the degree of g . For any integer n , note that

$$\begin{aligned} f(n) - g(n) &= \left(\frac{1}{e+1} n^d - b_e n^e \right) \\ &\quad + \left(\frac{1}{e+1} n^d - b_{e-1} n^{e-1} \right) \\ &\quad + \cdots + \left(\frac{1}{e+1} n^d - b_0 \right) \\ &\geq \left(\frac{1}{e+1} n^d - |b_e| n^e \right) \\ &\quad + \left(\frac{1}{e+1} n^d - |b_{e-1}| n^{e-1} \right) \end{aligned}$$

$$+ \cdots + \left(\frac{1}{e+1} n^d - |b_0| \right).$$

Since $d \geq e$, it follows that there exists an integer $n_0 \geq 1$ such that

$$\frac{1}{e+1} n^d - |b_e| n^e, \frac{1}{e+1} n^d - |b_{e-1}| n^{e-1}, \dots, \frac{1}{e+1} n^d - |b_0|$$

are positive for any $n \geq n_0$. This proves the Claim. $\square$

By the above Claim, it follows that $R(x)$ is the zero polynomial. This implies that

$$P(P(x) - 2015) = P(x)Q(x).$$

Since $P(x)$ is a nonconstant polynomial, it has a root z in $\mathbb{C}$. Substituting $x = z$ yields

$$P(-2015) = 0.$$

■

Example 1.5. Let $f(x), g(x)$ be polynomials with integer coefficients. Assume that for infinitely many integers n , $g(n)$ is nonzero and divides $f(n)$. Show that $g(x)$ divides $f(x)$ in the ring of polynomials with rational coefficients.

Solution 4. By division algorithm, there exist polynomials $q(x), r(x)$ with rational coefficients such that

$$f(x) = g(x)q(x) + r(x),$$

where $r(x)$ is the zero polynomial or has degree less than that of $g(x)$. Since $q(x), r(x)$ have rational coefficients, there exists a positive integer m such that $mq(x), mr(x)$ have integer coefficients. Note that

$$mf(x) = g(x)(mq(x)) + mr(x).$$

For infinitely many integers n , we have $g(n) \neq 0$ and $g(n)$ divides $mr(n)$. Let us consider the case that $r(x)$ is not the zero polynomial. Since $r(x)$ has degree less than that of $g(x)$, there exists a positive integer N such that

$$|mr(n)| < |g(n)|$$

for all integers $n > N$. It follows that $mr(n) = 0$ for infinitely many integers n . Hence, $r(x)$ is the zero polynomial, which is a contradiction. Therefore, $r(x)$ is the zero polynomial. Since $f(x), g(x)$ have integer coefficients, and $g(x)$ divides $f(x)$ in the ring of polynomials with rational coefficients. $\blacksquare$

References

[Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)